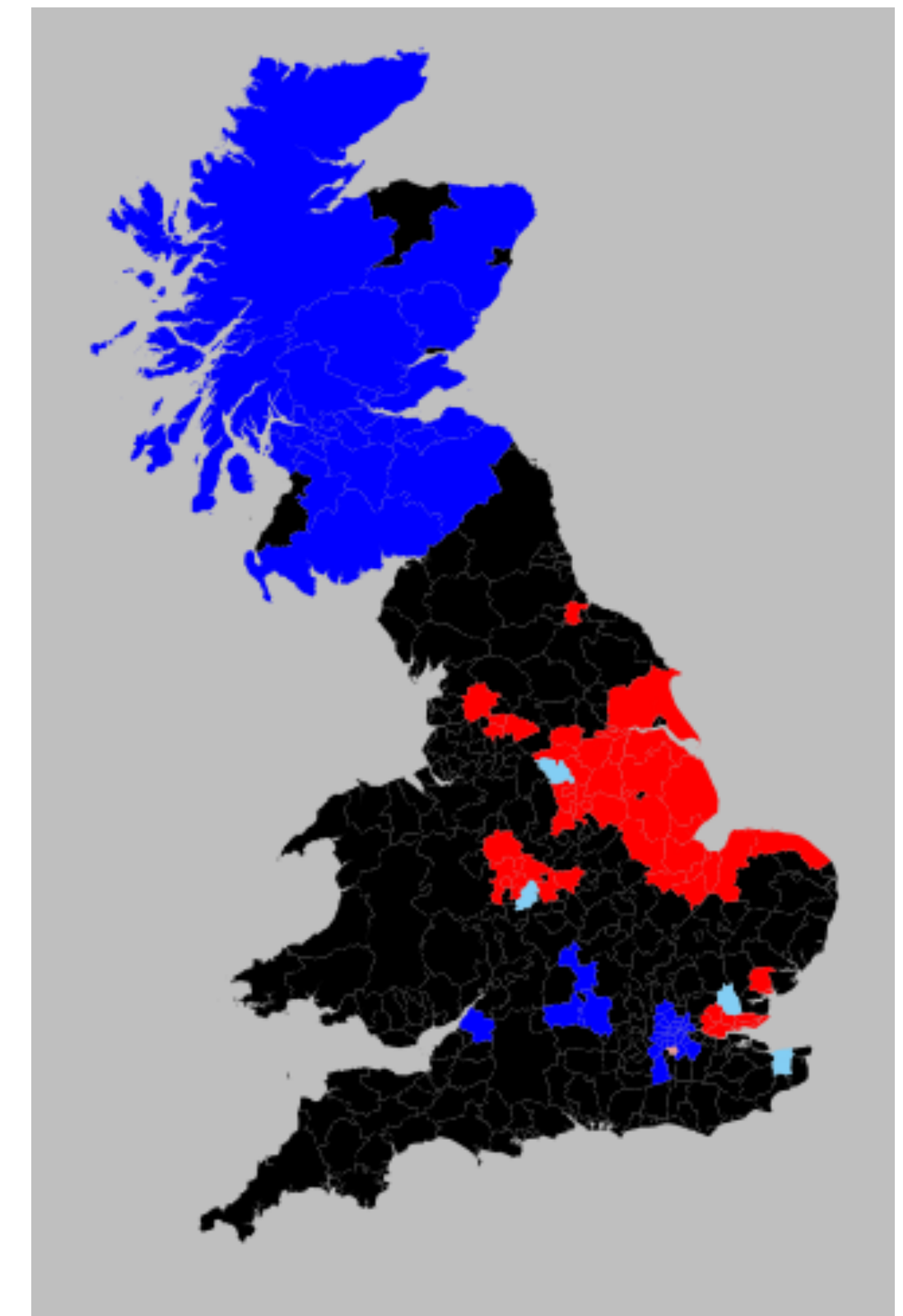


Lecture 6: Spatial autocorrelation

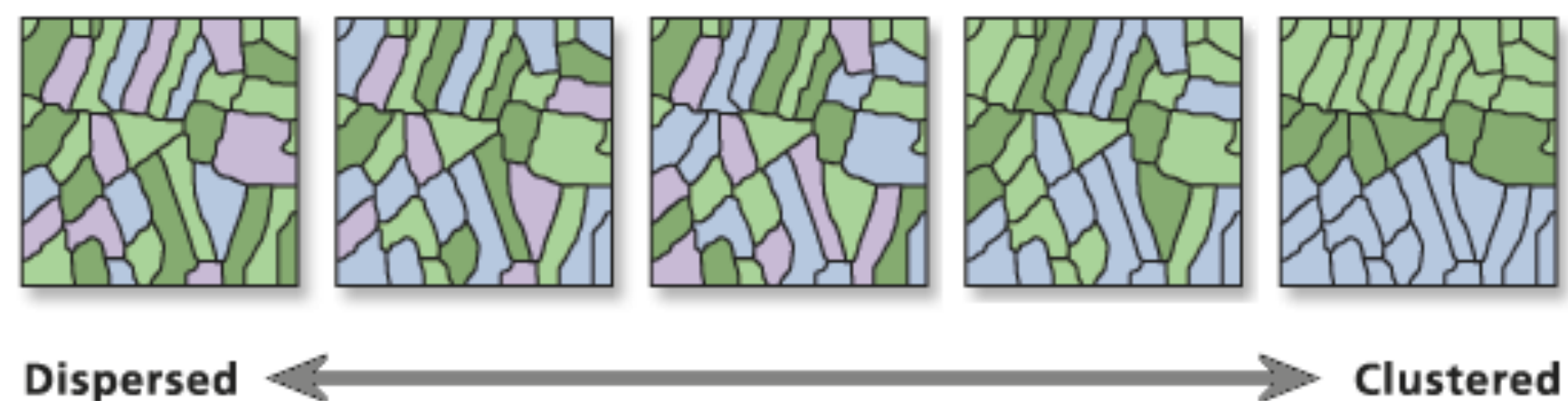
Instructor: Michael Szell

Mar 4, 2026

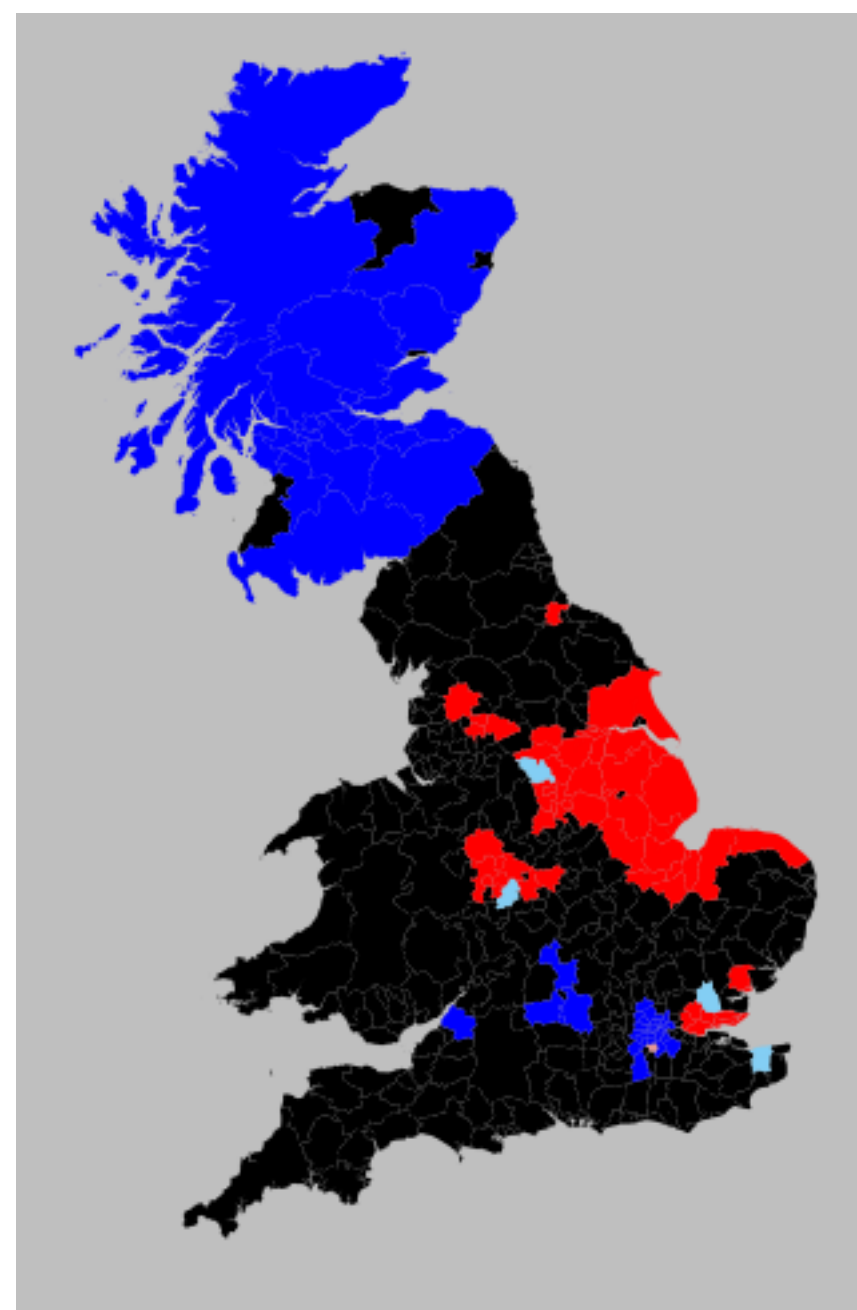


Today you will learn about spatial correlation

Questions to ask in
Exploratory spatial
data analysis



Use in Python
with example



Global and local
measures

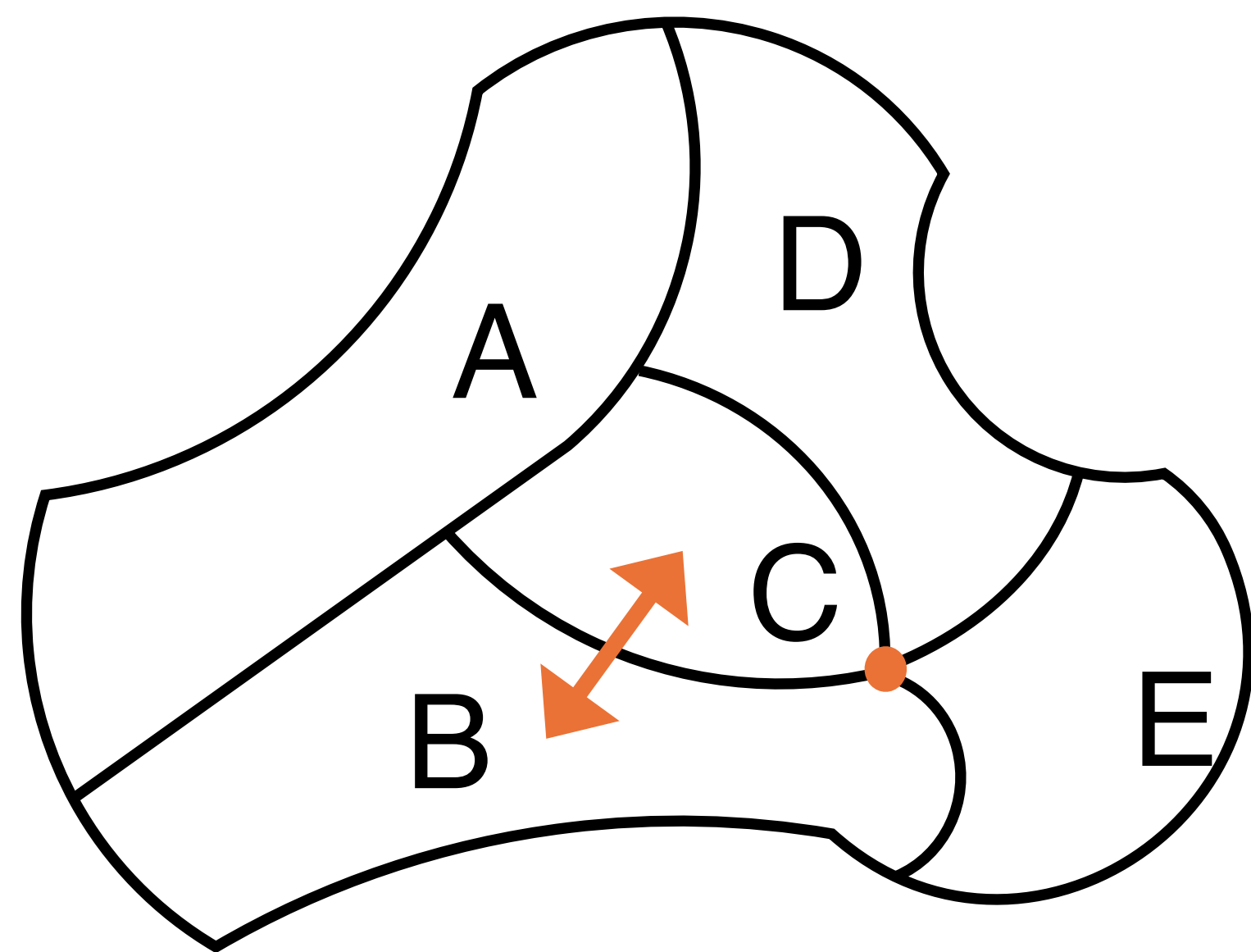
$$I = \frac{n}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij} z_i z_j}{\sum_i z_i^2}$$

$$I_i = \frac{z_i}{m_2} \sum_j w_{ij} z_j ; m_2 = \frac{\sum_i z_i^2}{n}$$

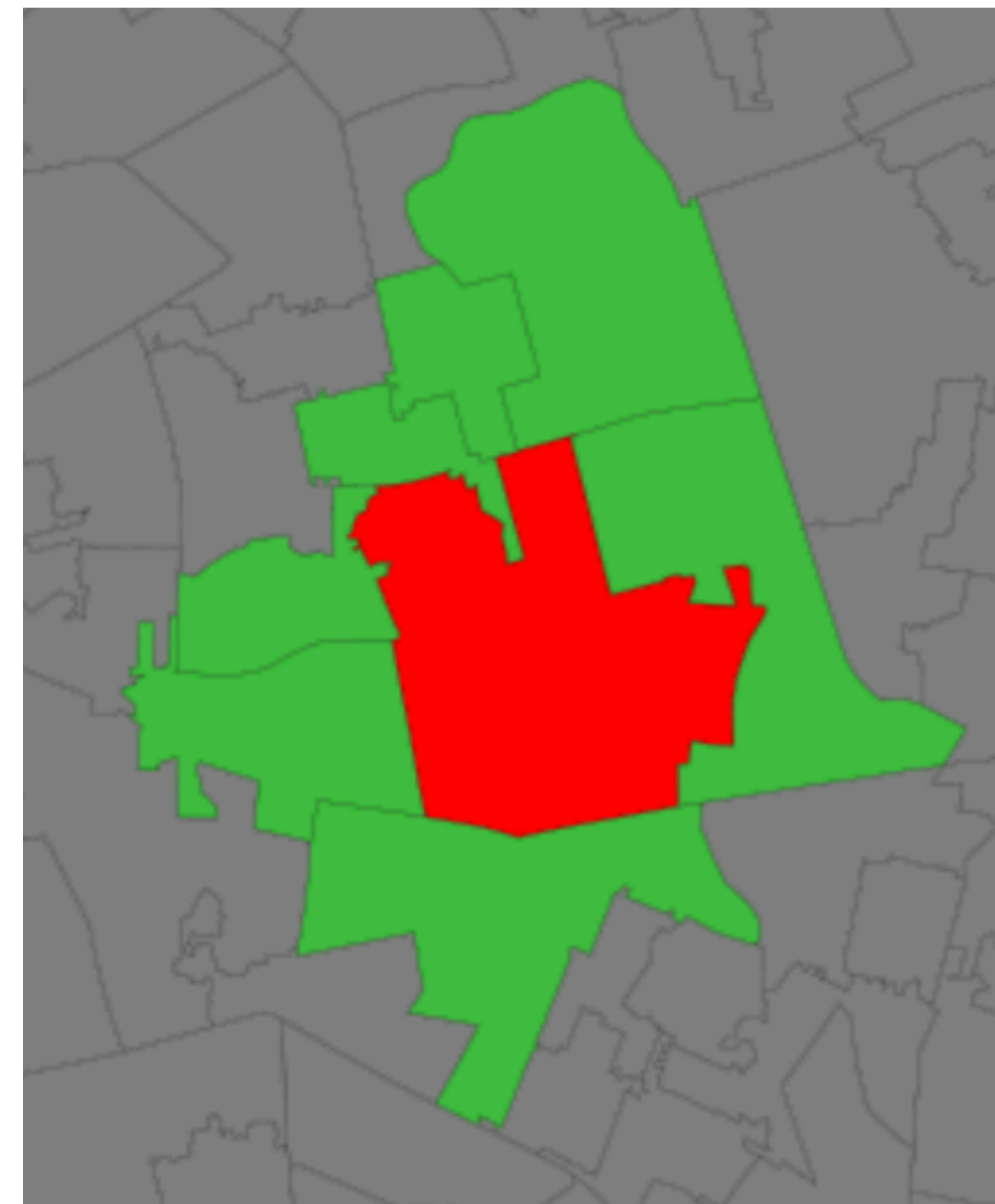
Exam project info



Recap: Spatial weights encodes information about how objects are related to each other in space



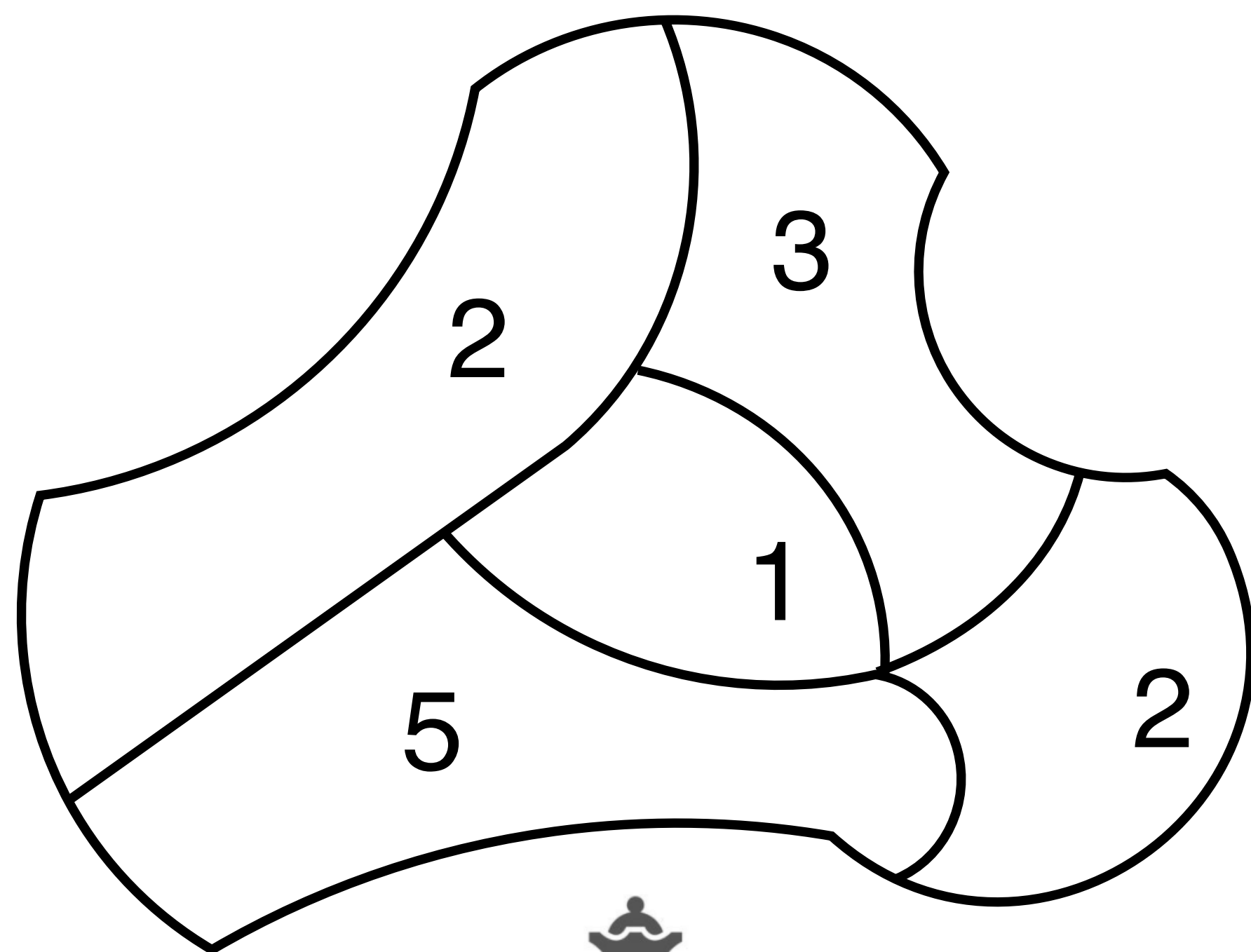
$$W = \begin{pmatrix} 0 & w_{12} & \dots & w_{1N} \\ w_{21} & \ddots & w_{ij} & \vdots \\ \vdots & w_{ji} & 0 & \vdots \\ w_{N1} & \dots & \dots & 0 \end{pmatrix}$$



The spatial lag is the weighted average value of neighbors

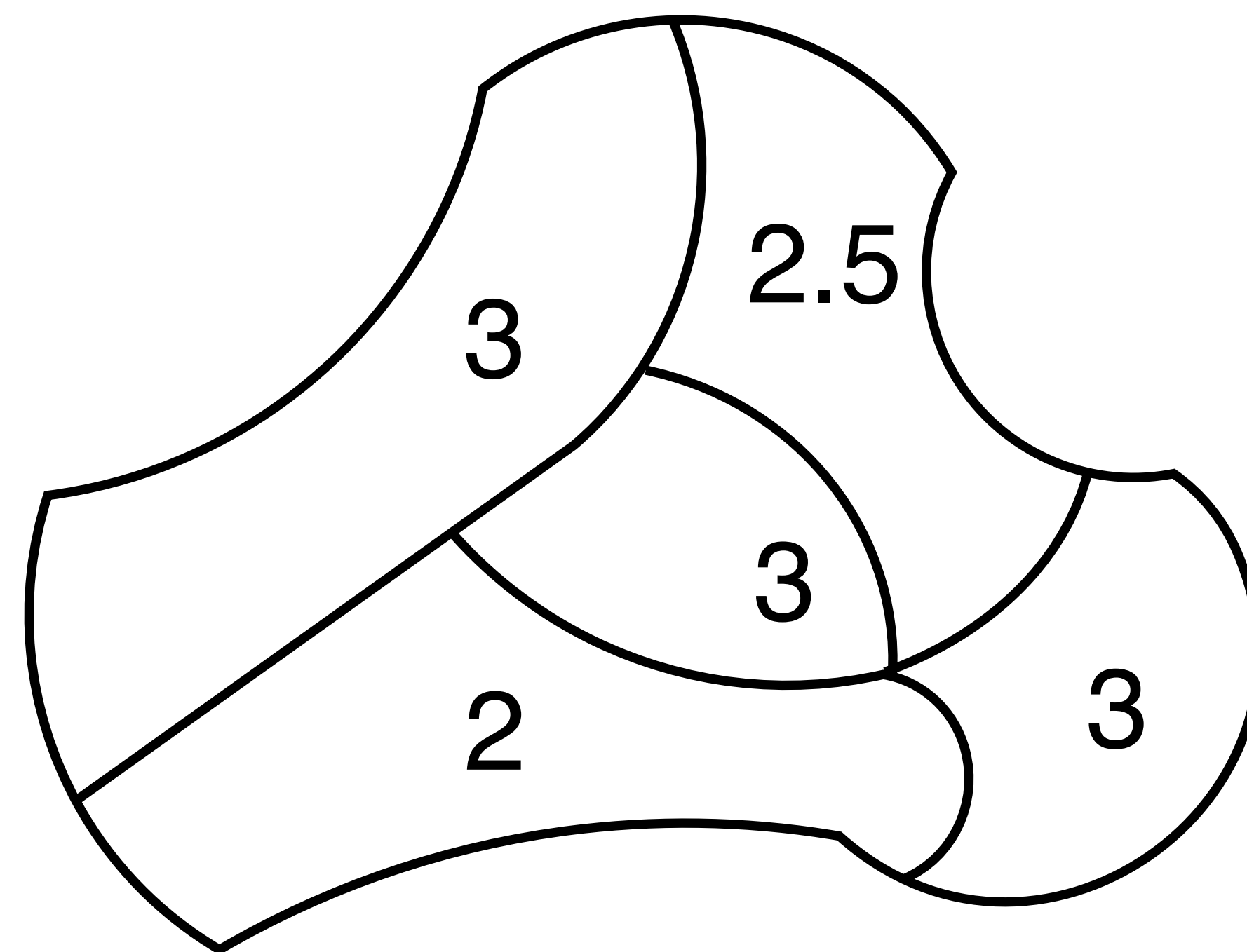
if W is standardized

Values y



QUEEN

Spatial lag y_{lag}



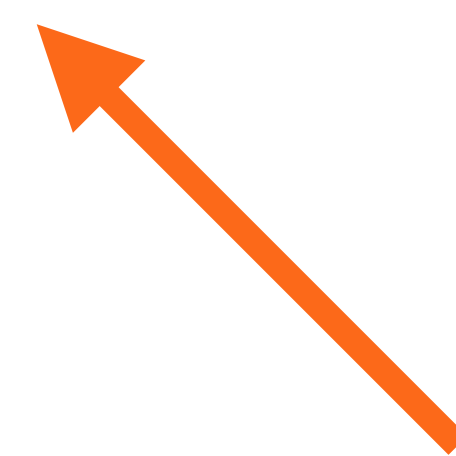
It is a smoother: It brings all values closer to the average

GDS asks: How do things relate in space?

Everything is related to everything else,
but near things are more related than
distant things.

Tobler's 1st law of geography

ESDA: Exploratory Spatial Data Analysis



Assumption-free investigation of spatial patterns with statistical techniques

Is a variable concentrated over space?
Do similar values locate closeby? If so, where?
Given a choropleth map, is there a significant statistical pattern?

Is a variable concentrated over space?
Do similar values locate closeby? If so, where?
Given a choropleth map, is there a significant statistical pattern?

What process is behind a pattern?
Why are there certain clusters?

Spatial autocorrelation describes the
relationship between values and
locations

Spatial autocorrelation: Positive vs. Negative

Is the spatial counterpart of traditional correlation

Positive

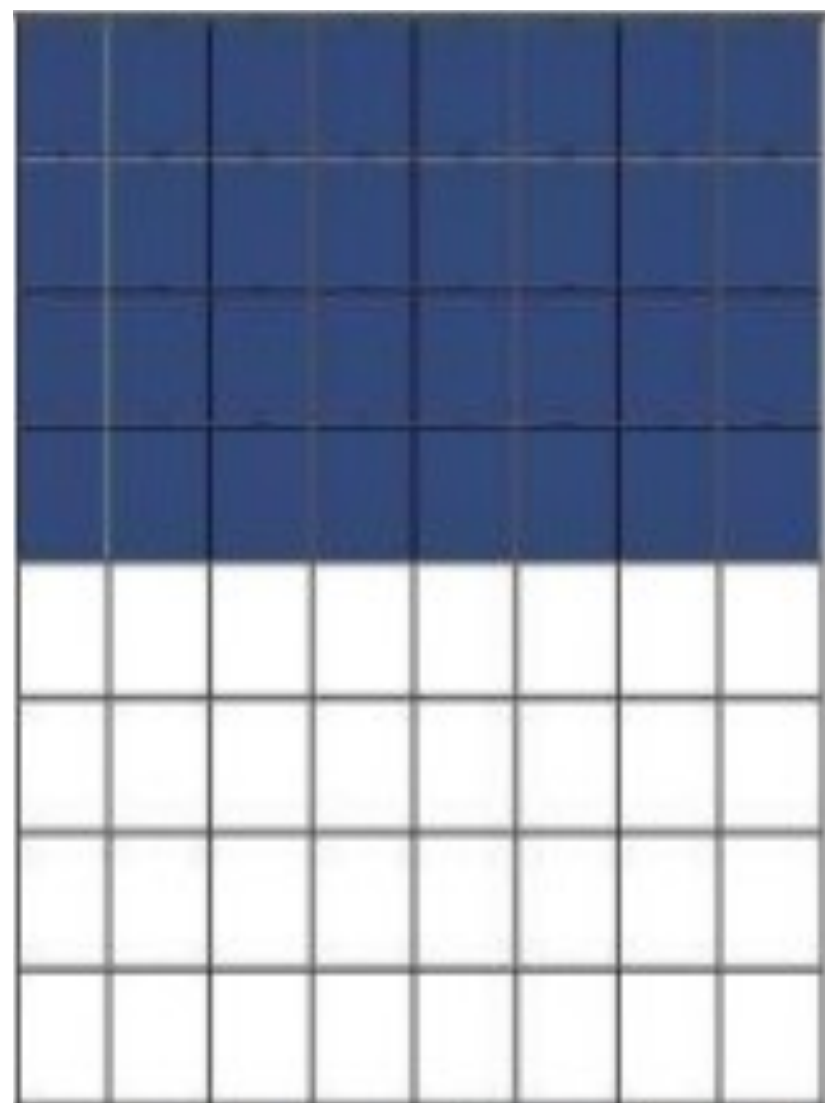
Negative

Spatial autocorrelation: Positive vs. Negative

Is the spatial counterpart of traditional correlation

Positive

similar values
are closeby



Negative

similar values
are further apart

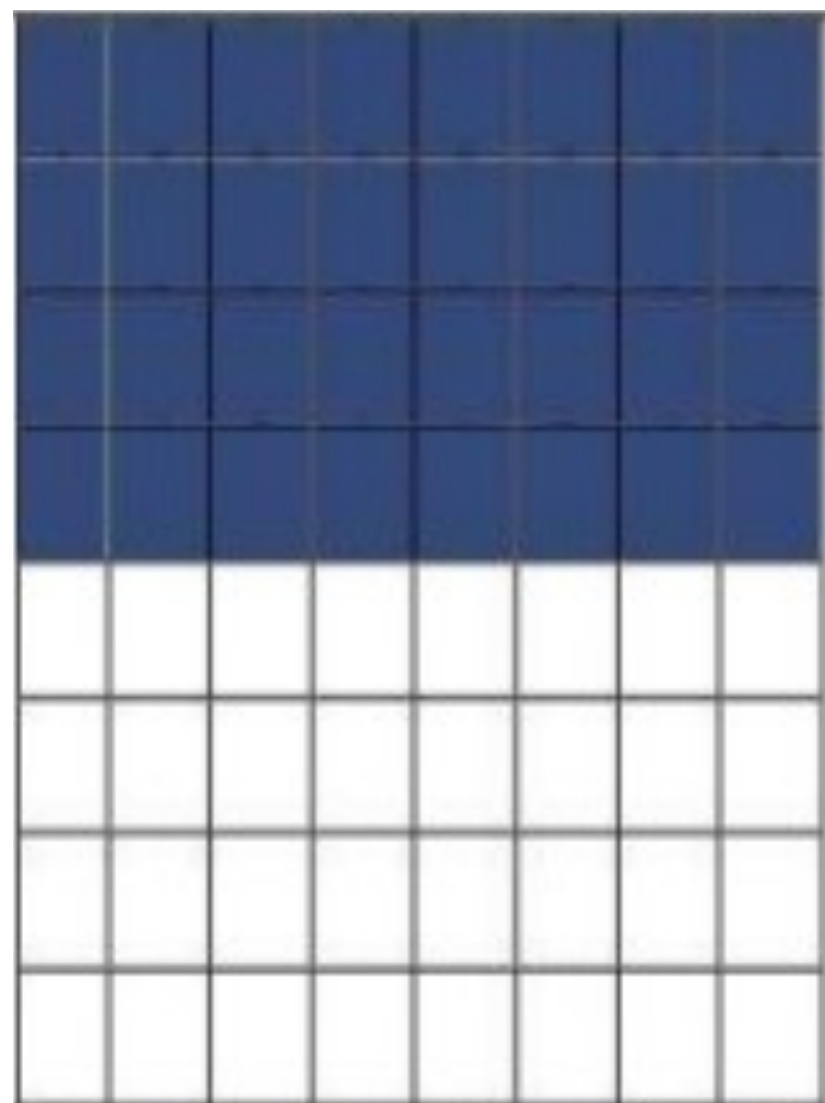


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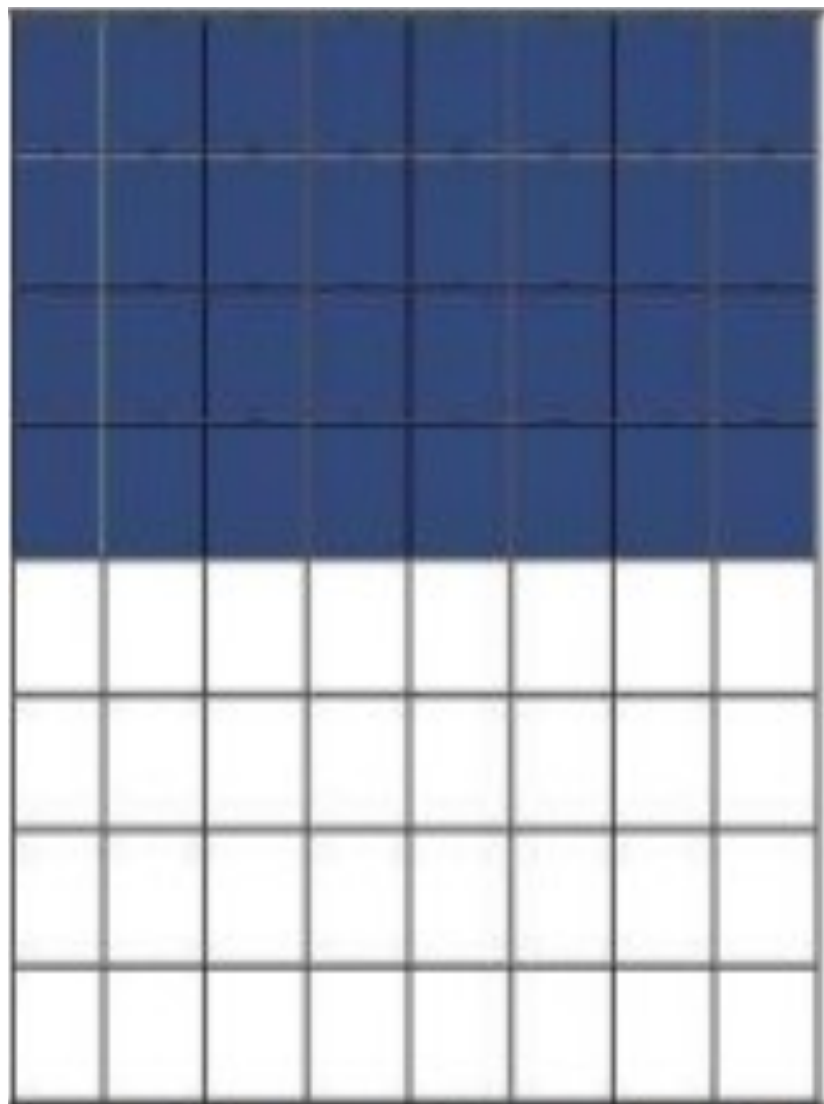


Spatial autocorrelation: Positive vs. Negative

Is the spatial counterpart of traditional correlation

Positive

similar values
are closeby



income, poverty, covid cases,
vegetation, temperature,...

Negative

similar values
are further apart



tigers, pharmacies, fire/police/
metro stations, hospitals,...

WHY DO COMPETITORS OPEN THEIR STORES NEXT TO ONE ANOTHER?

TEDEd

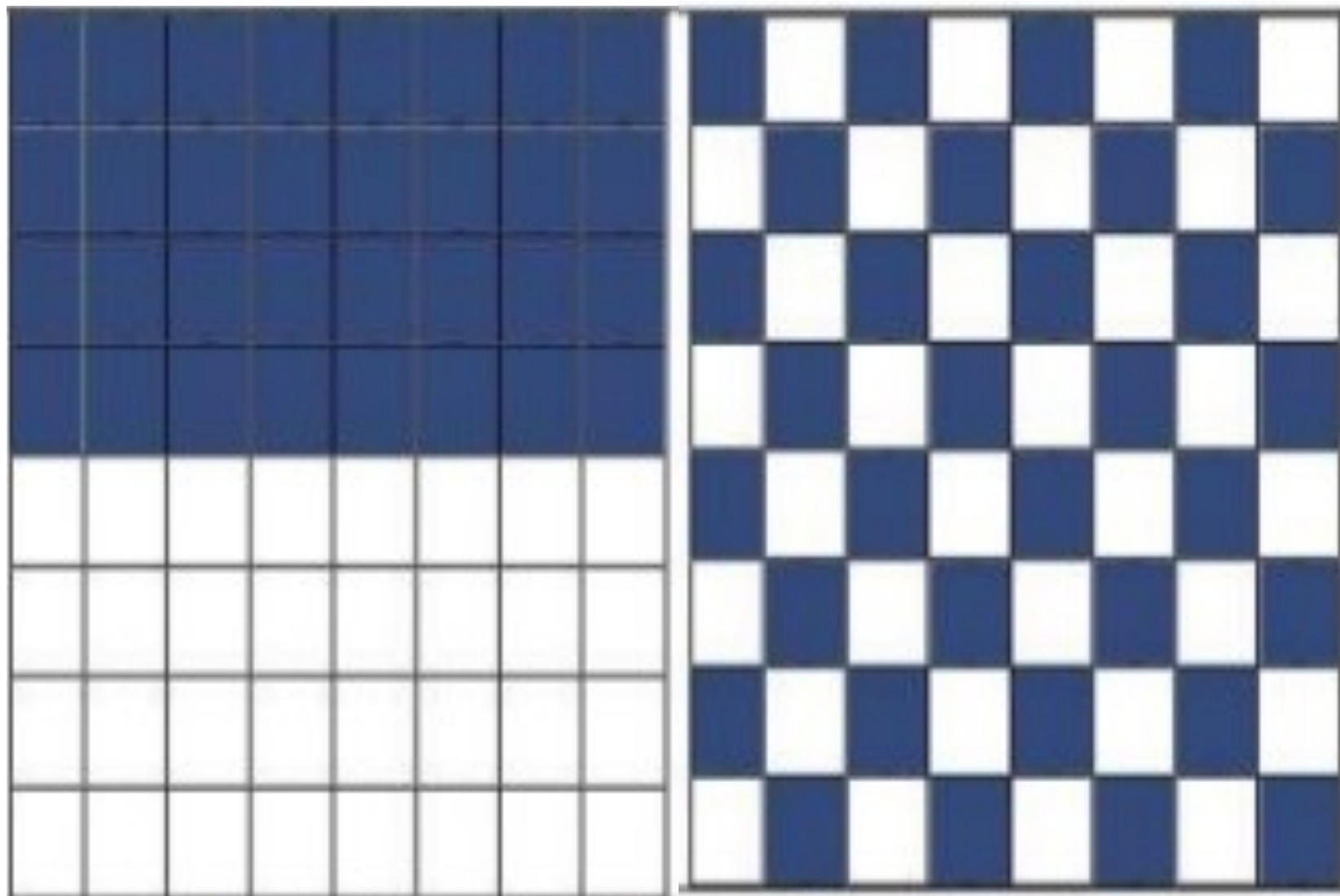


Spatial autocorrelation: Global vs. Local

Is the spatial counterpart of traditional correlation

Global

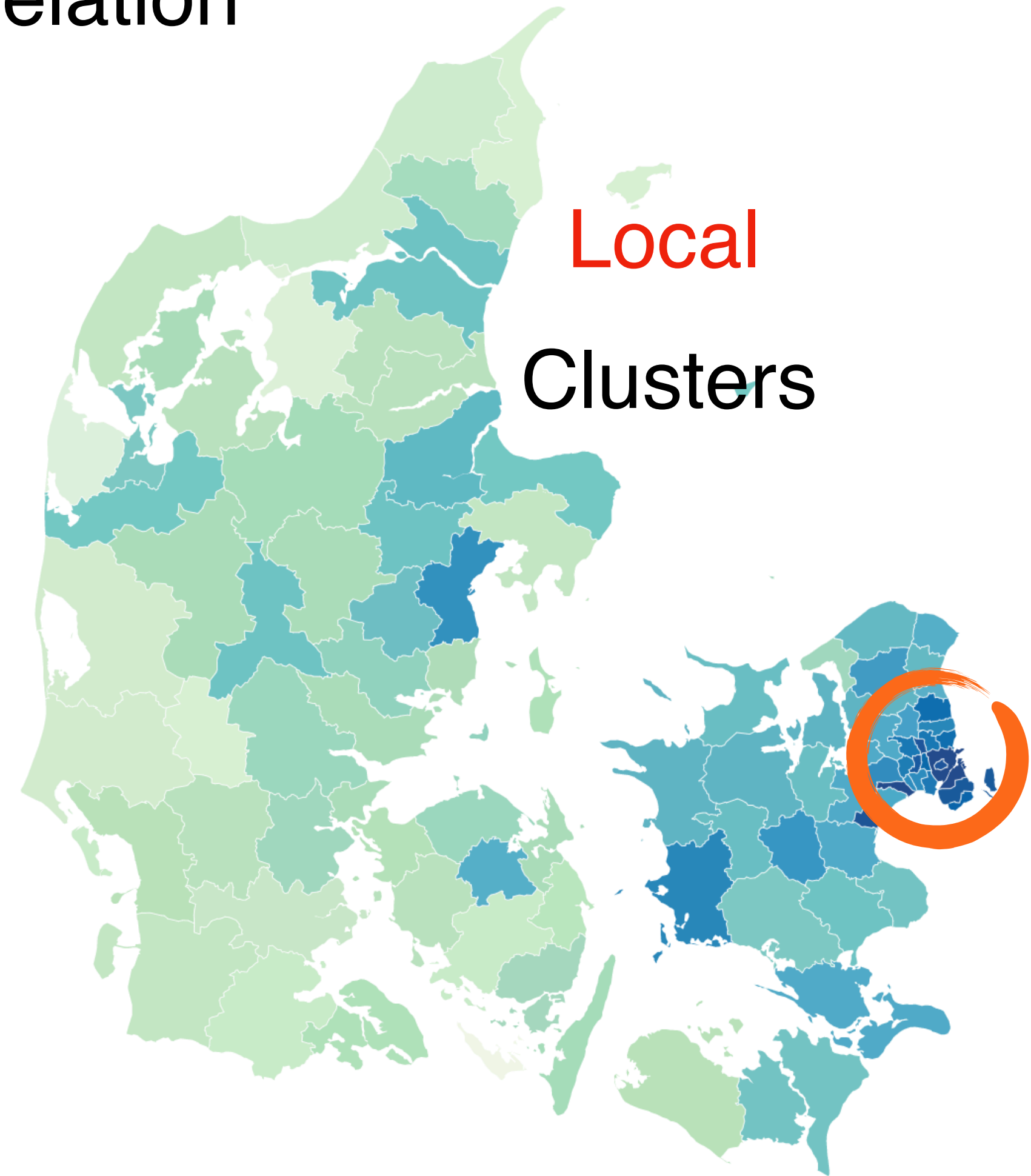
Clustering



Do values tend to be close to (dis)similar values?

Local

Clusters

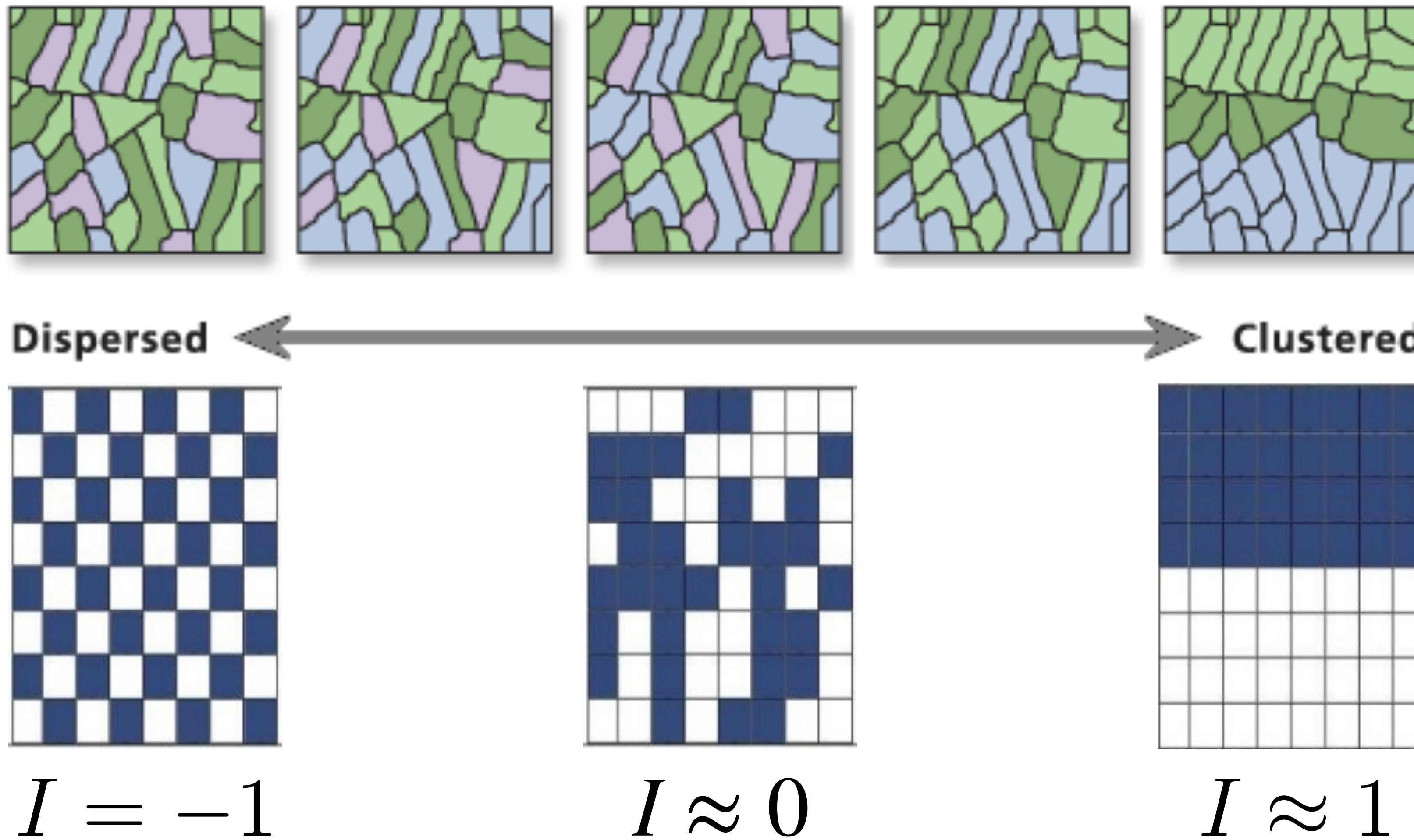


Are there areas with an extraordinary concentration of (dis)similar values?

Global Spatial Autocorrelation

Global spatial autocorrelation: Moran's I

Moran's I measures the average correlation between the value of a variable at one location and the value at nearby locations.



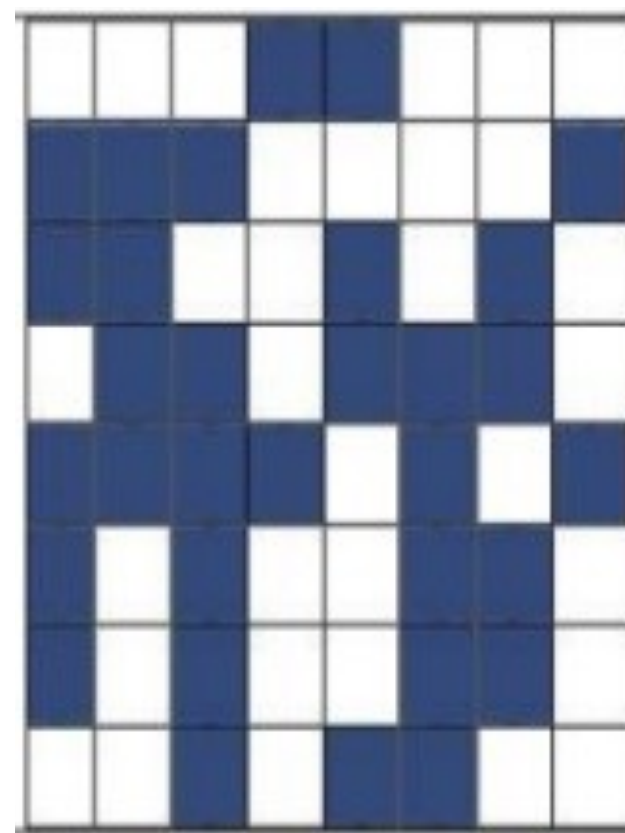
Global spatial autocorrelation: Moran's I

Moran's I measures the average correlation between the value of a variable at one location and the value at nearby locations.

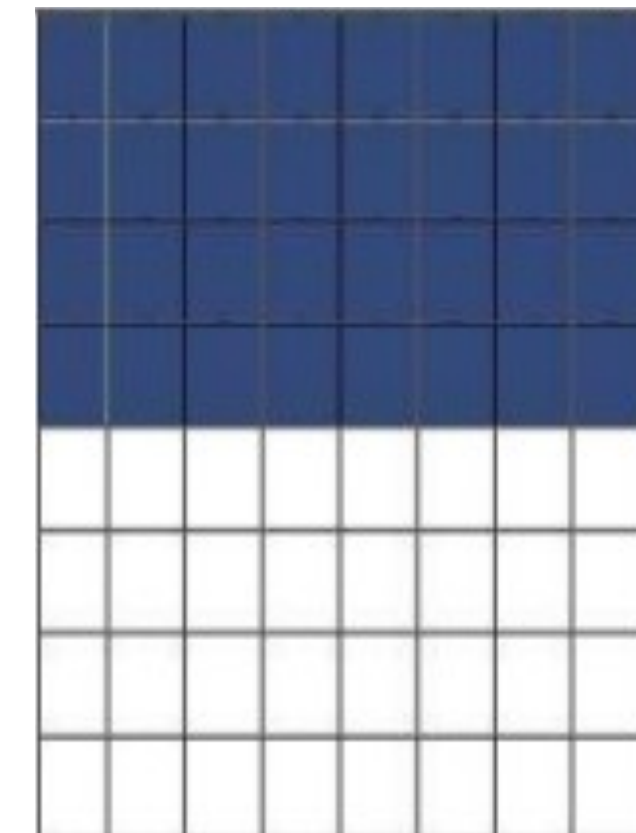
Statistical inference with a significance value:
How likely is it to obtain a map like observed from a purely random pattern?



$$I = -1$$



$$I \approx 0$$



$$I \approx 1$$

Global spatial autocorrelation: Moran's I

Moran's I measures the average correlation between the value of a variable at one location and the value at nearby locations.

$$y_i$$

$$y_{\text{lag},i} = \sum_{j=1}^n w_{ij} y_j$$

$$\rho = \frac{1}{n-1} \sum \left(\frac{y_i - \bar{y}}{s_y} \right) \left(\frac{y_{\text{lag},i} - \bar{y}_{\text{lag}}}{s_{y_{\text{lag}}}} \right)$$

Global spatial autocorrelation: Moran's I

Moran's I measures the average correlation between the value of a variable at one location and the value at nearby locations.

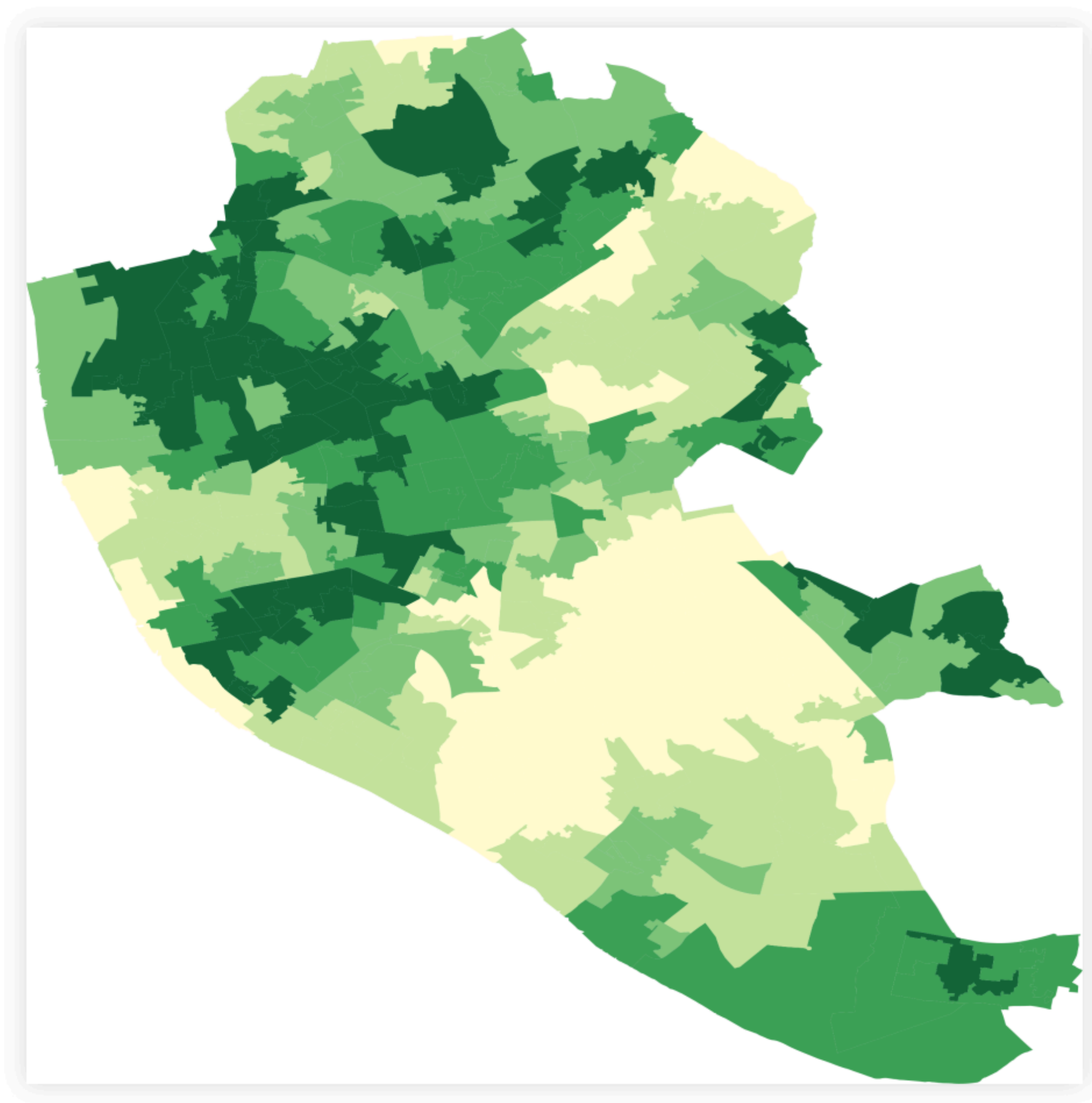
$$y_i$$

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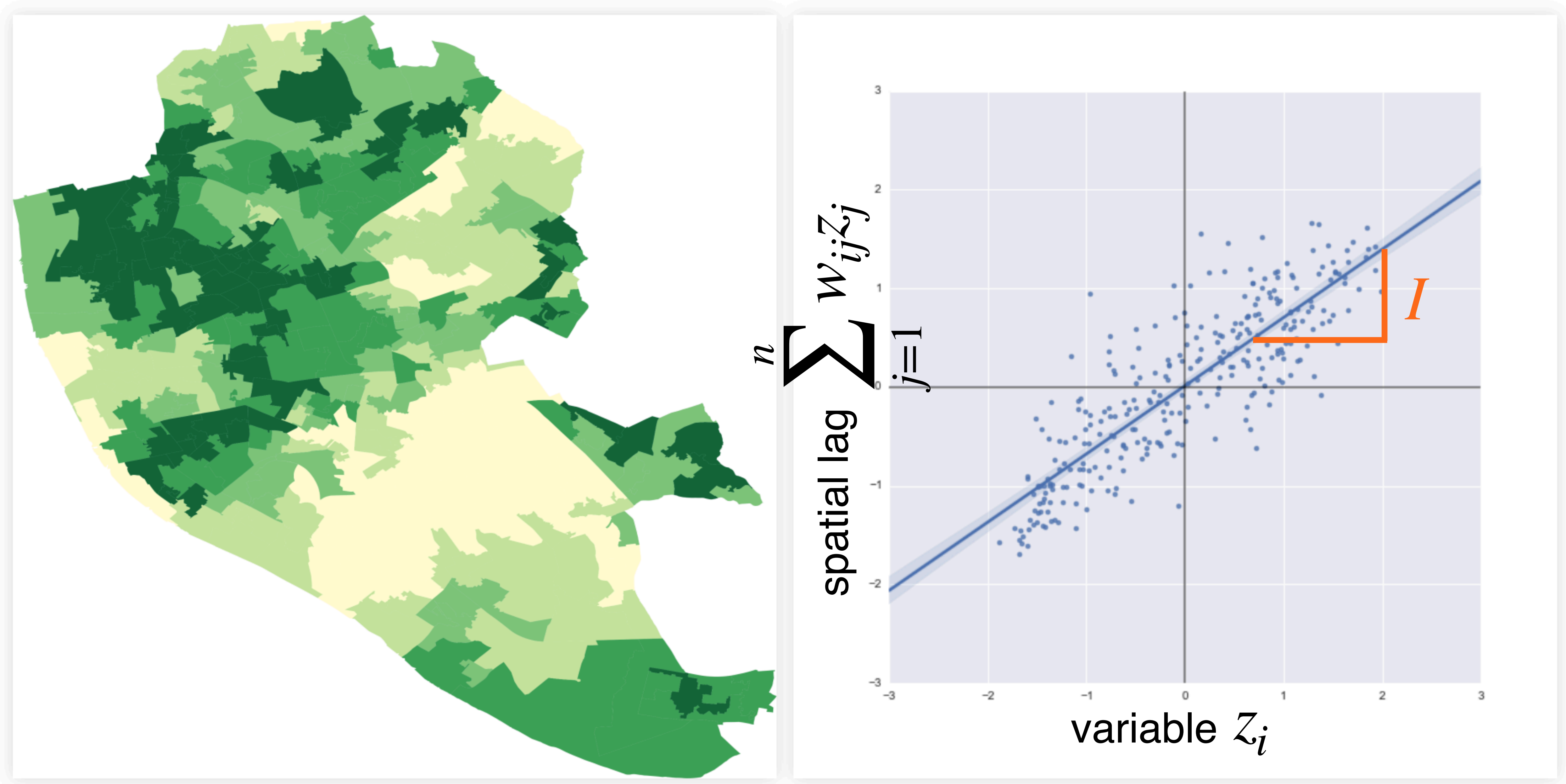
$$\rho = \frac{1}{n-1} \sum \left(\frac{y_i - \bar{y}}{s_y} \right) \left(\frac{y_{\text{lag},i} - \bar{y}_{\text{lag}}}{s_{y_{\text{lag}}}} \right)$$

If we call $z_i = \left(\frac{y_i - \bar{y}}{s_y} \right)$, then:

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij} z_i z_j}{\sum_i z_i^2}$$



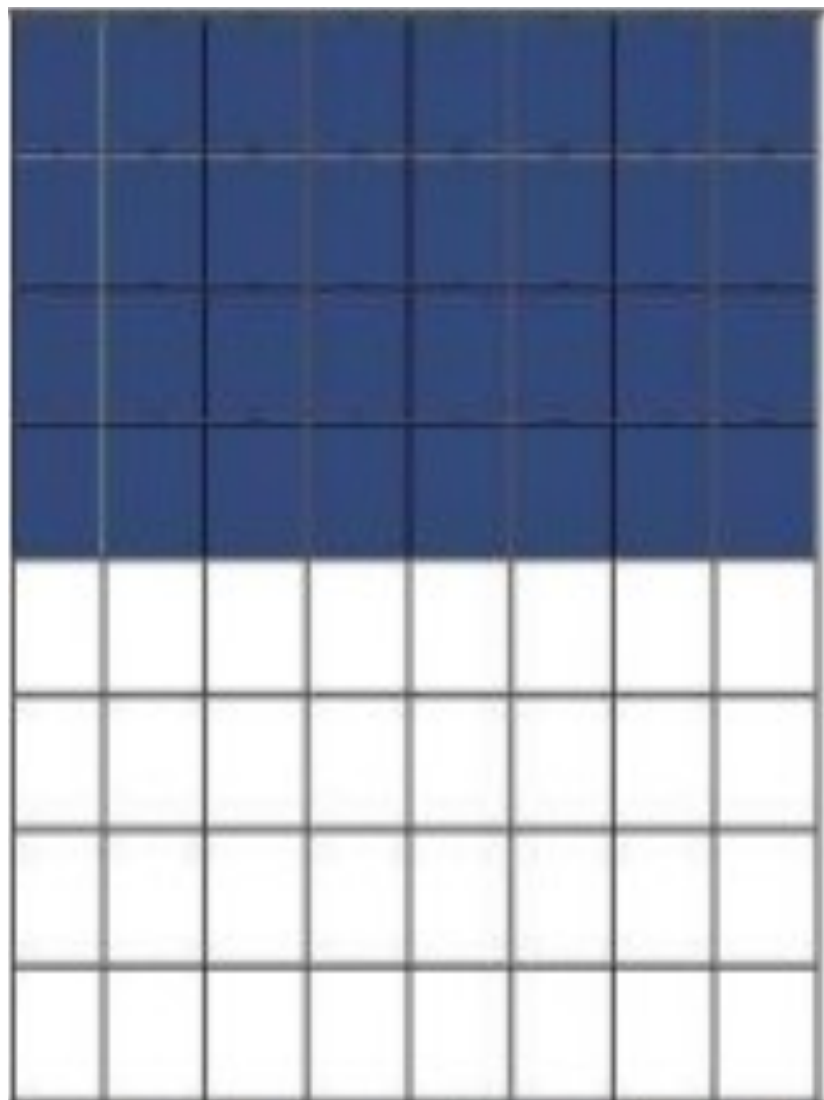
Moran's I is the regression slope in the **Moran plot**



Random distribution is our null hypothesis

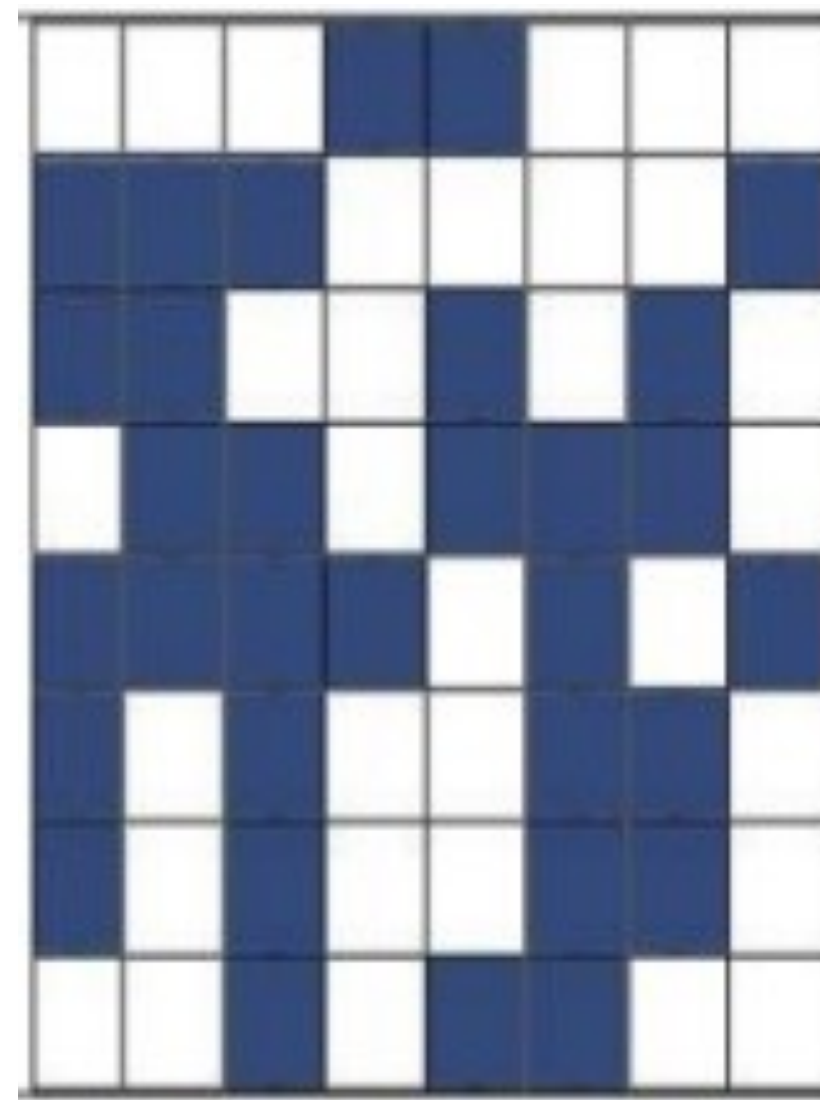
Positive

similar values
are closeby



No

values are
randomly distributed

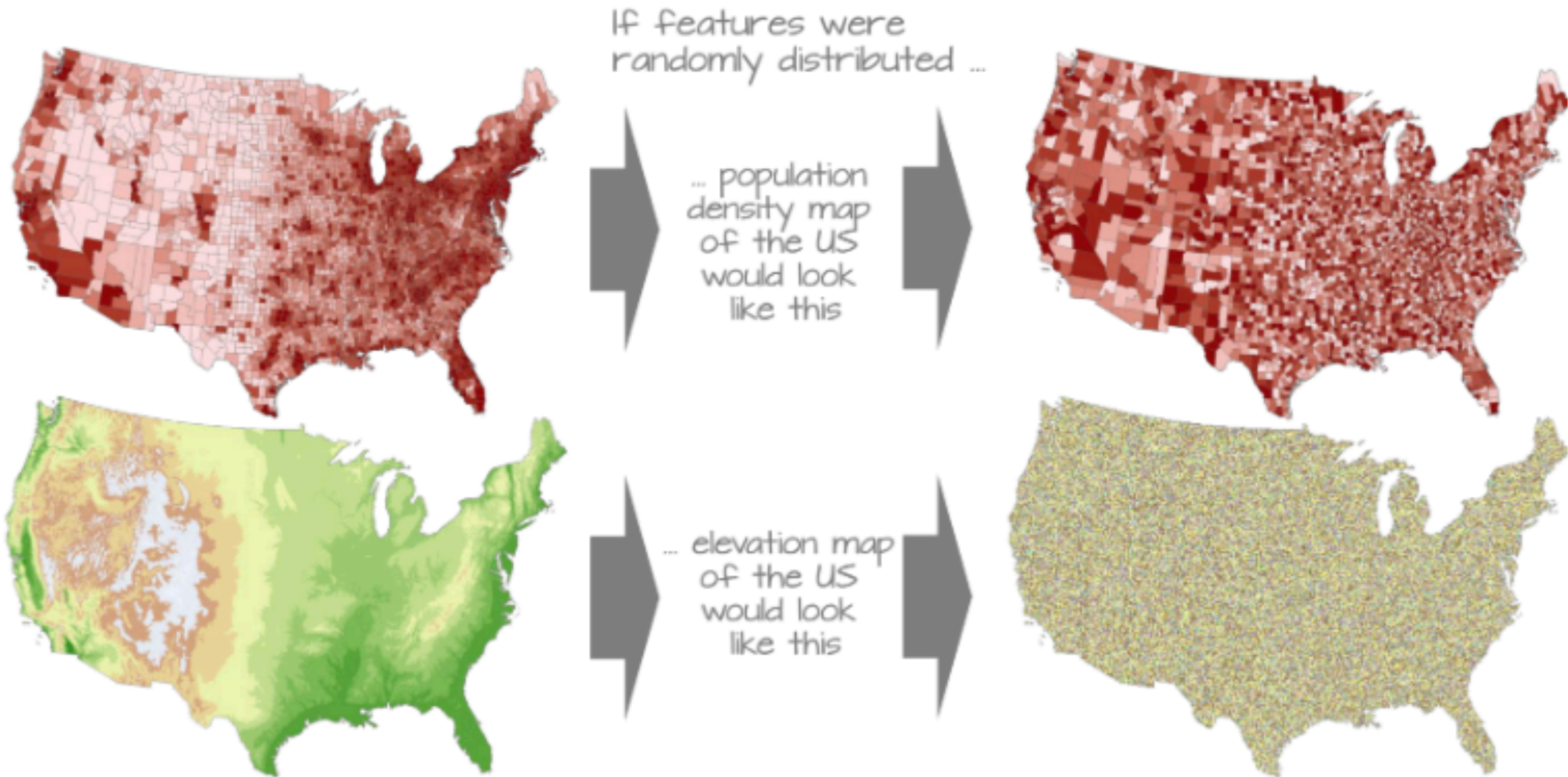


Negative

similar values
are further apart



Random distribution is our null hypothesis



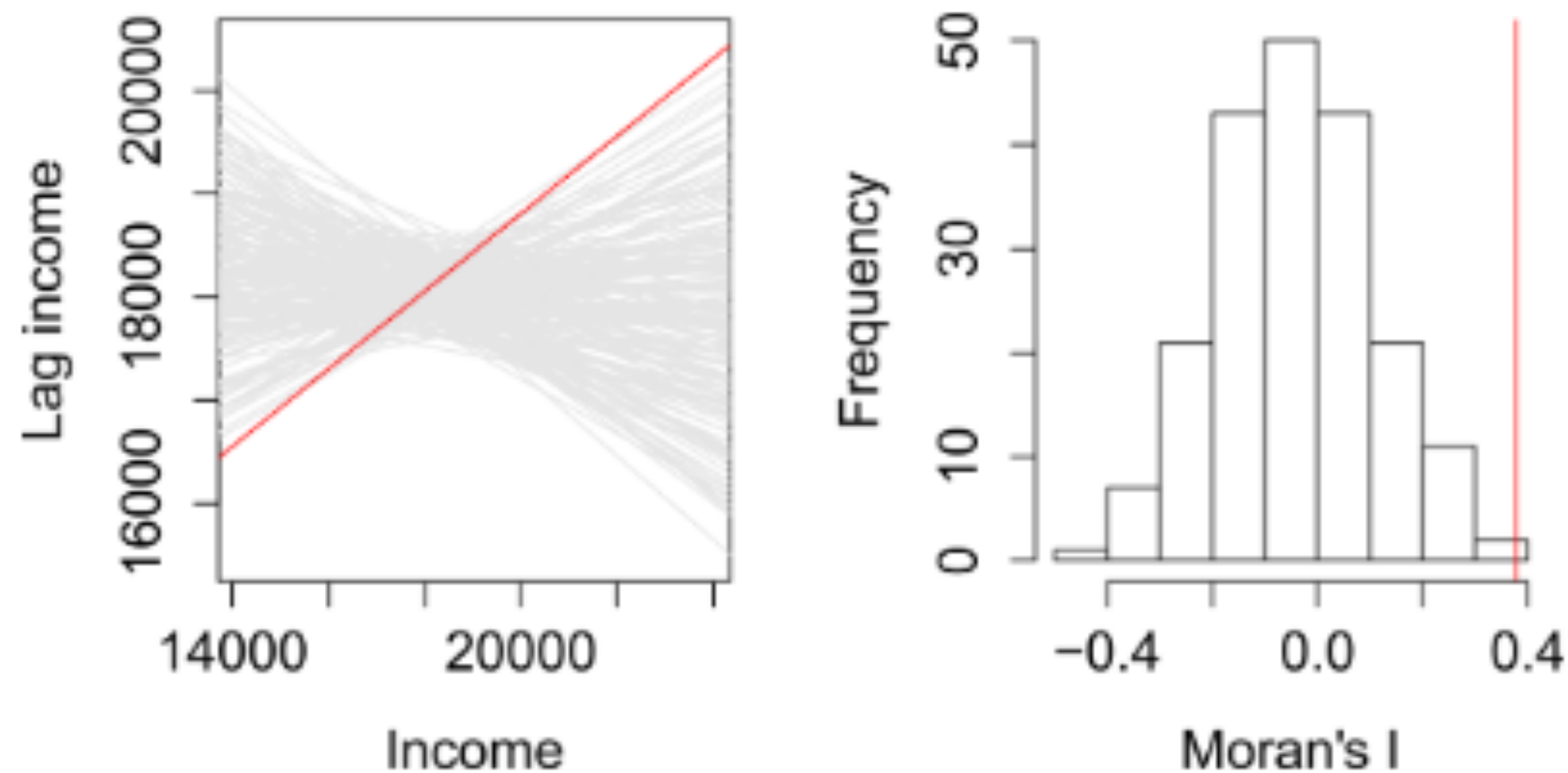
We need to test whether the pattern is significant

Null hypothesis is a random pattern



We need to test whether the pattern is significant

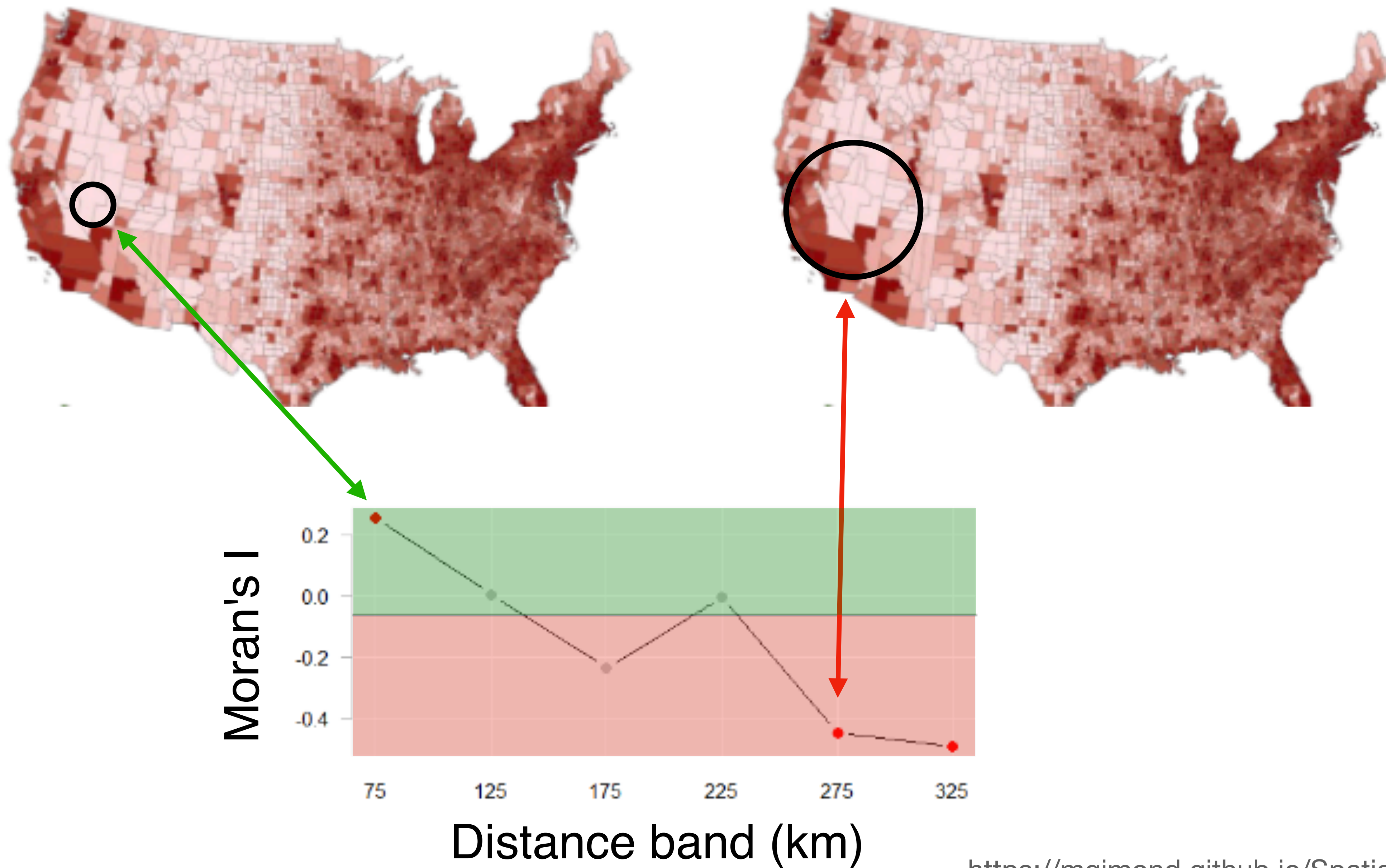
Null hypothesis is a random pattern



$$P_{pseudo} = \frac{N_{extreme} + 1}{N + 1}$$



Moran's I can depend on scale (distance band neighborhood)



Local Spatial Autocorrelation

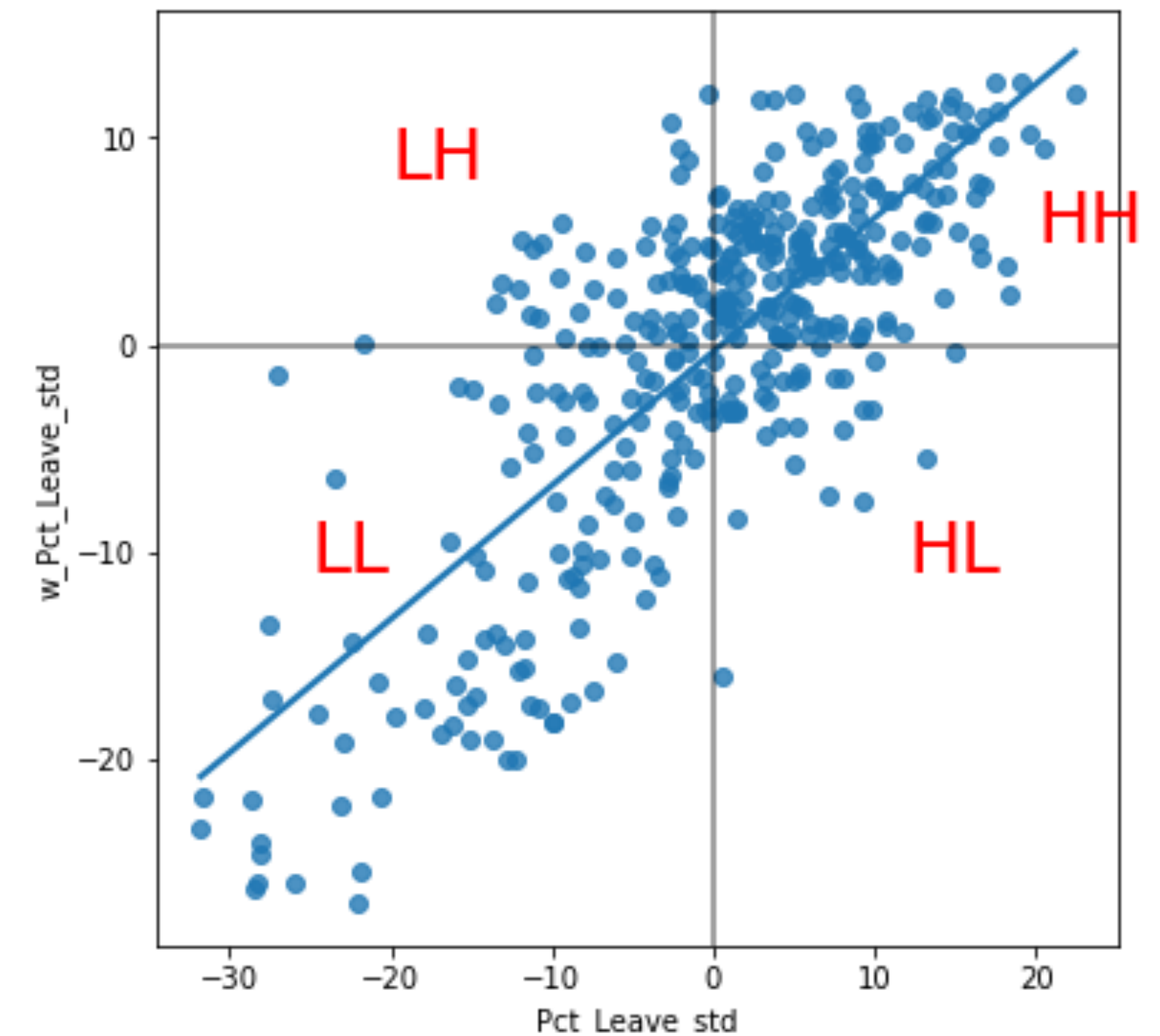
Local spatial autocorrelation is about local clusters

Cluster = Portion of a map where values are correlated in a particularly strong or specific way

Local spatial autocorrelation is about local clusters

Cluster = Portion of a map where values are correlated in a particularly strong or specific way

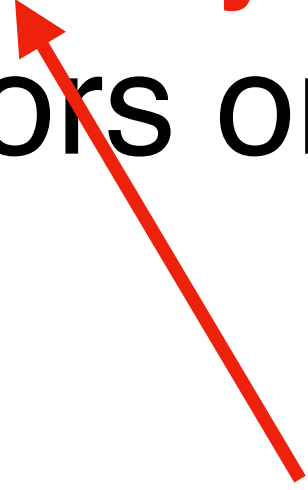
Positive	High-High: Hotspot
	Low-Low: Coldspot
Negative	High-Low: Spatial outlier
	Low-High: Spatial outlier (donut)



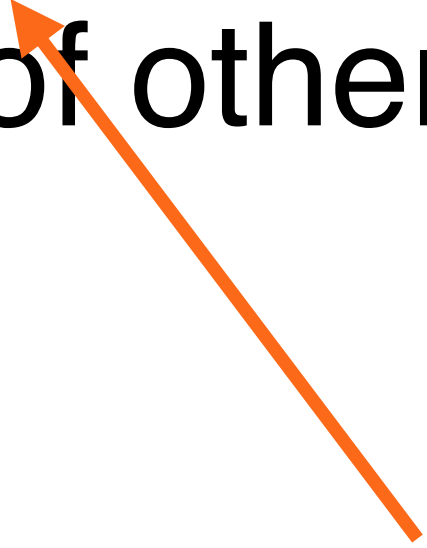
LISA: Local Indicators of Spatial Association

To what extent is **each object's relation with its neighbors** **significantly** different from the relations of other objects and their neighbors on the map?

2. Test for significance

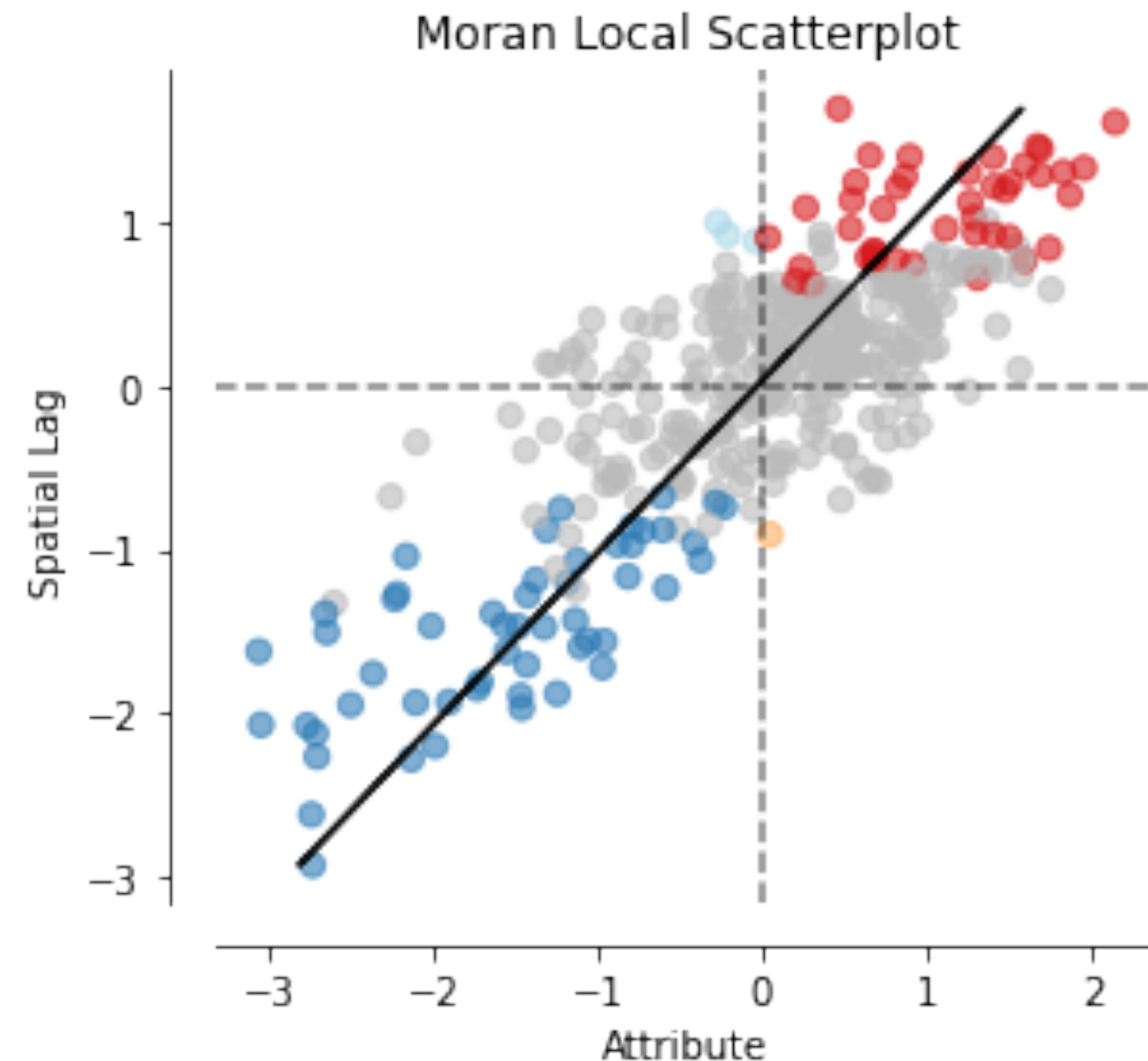


1. Test for spatial cluster



LISA: Local Indicators of Spatial Association

To what extent is **each object's relation with its neighbors** **significantly** different from the relations of other objects and their neighbors on the map?



LISA: Local Moran's I

$$I_i = \frac{z_i}{m_2} \sum_j w_{ij} z_j$$

$$m_2 = \frac{\sum_i z_i^2}{n}$$

$$z_i = y_i - \bar{y}$$

LISA: Local Moran's I

$$I_i = \frac{z_i}{m_2} \sum_j w_{ij} z_j \quad m_2 = \frac{\sum_i z_i^2}{n} \quad z_i = y_i - \bar{y}$$

If W is row-standardized, then:

$$\sum_i I_i = \gamma I$$

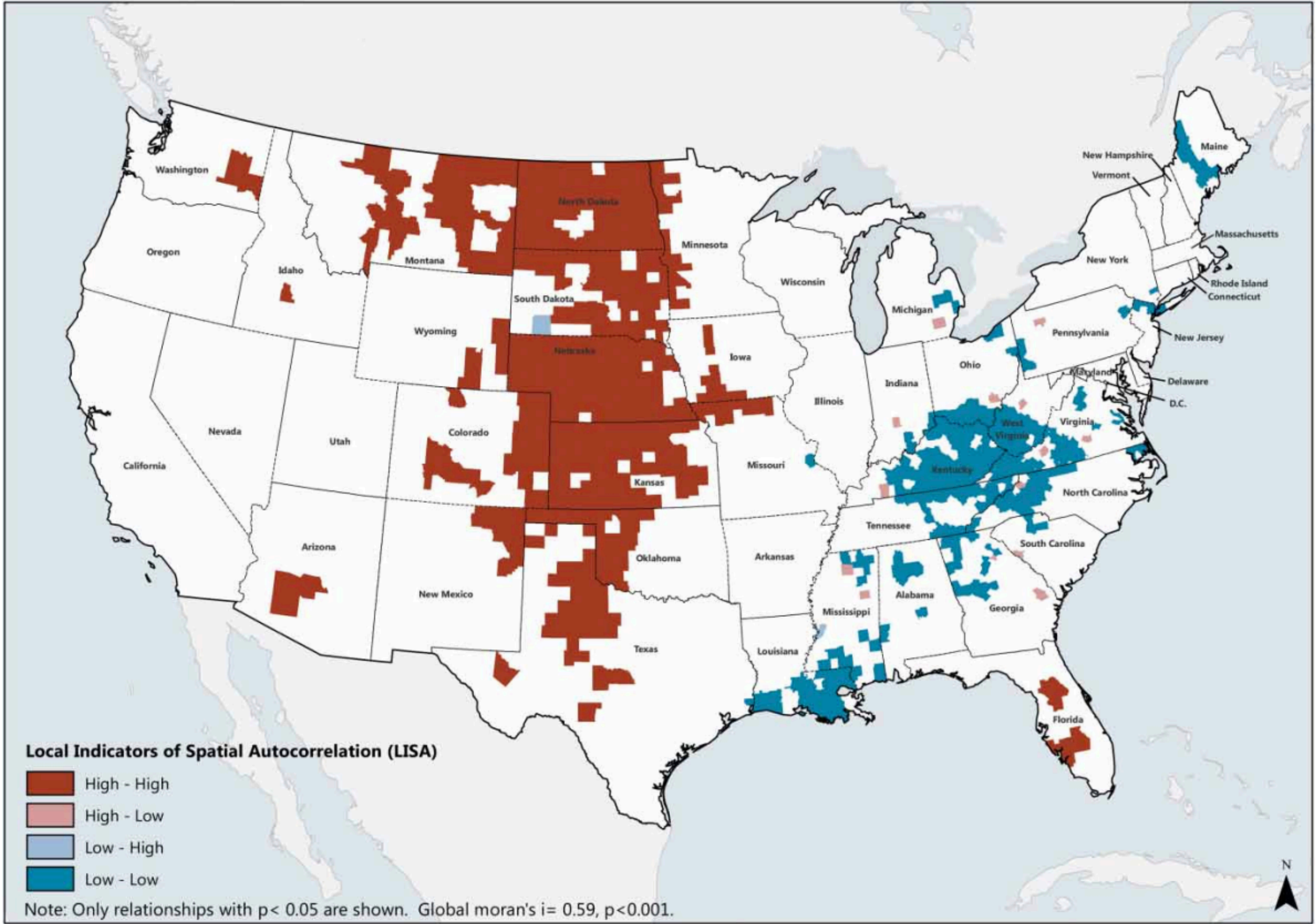
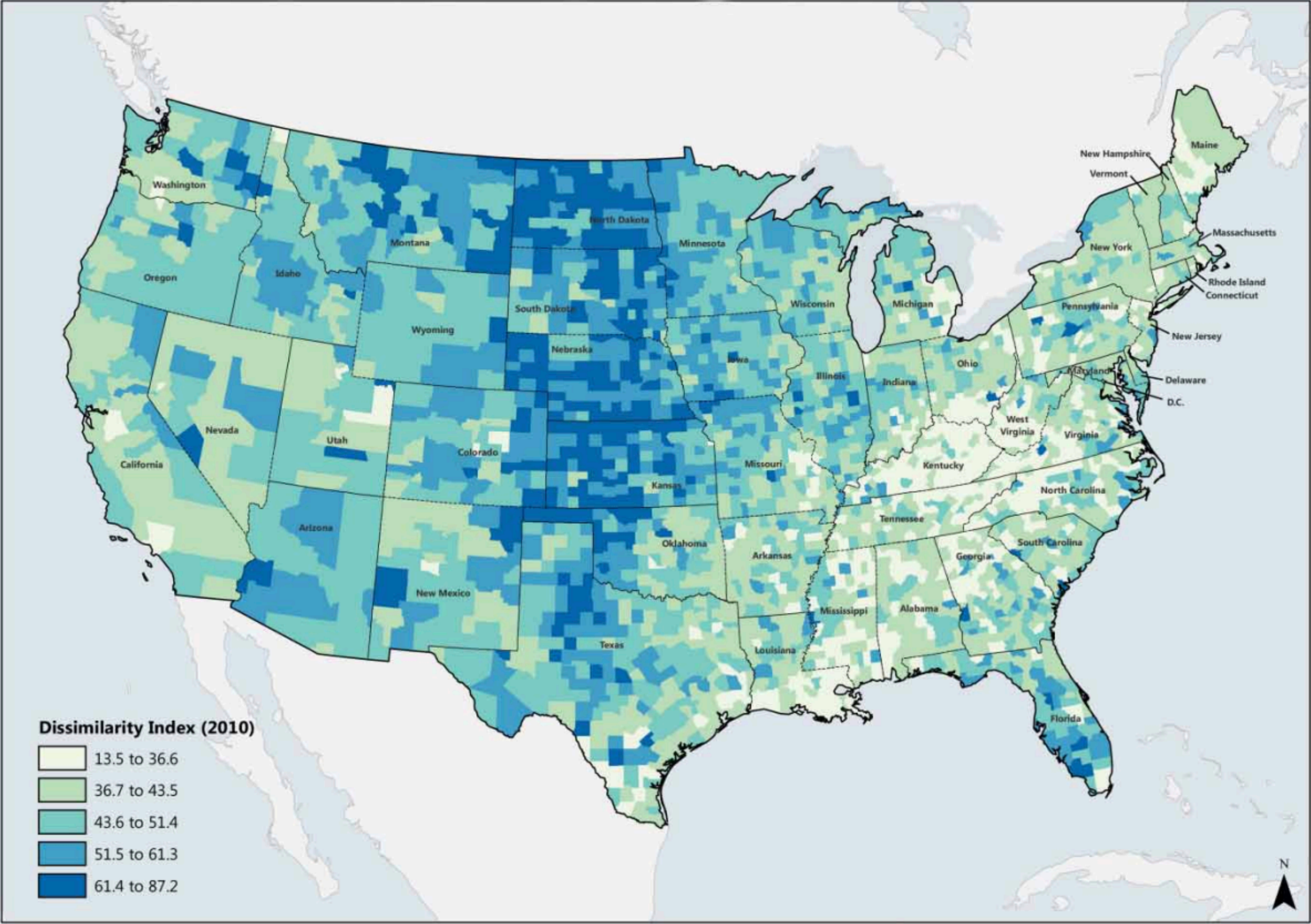
$$\gamma = \sum_i \sum_j w_{ij} = \text{scale factor}$$

$I =$ global indicator of autocorrelation

= Sum of local indicators proportional to global indicator

Applications of LISA: Spatial processes

Residential segregation between young and old



Applications of LISA: significant regional differences

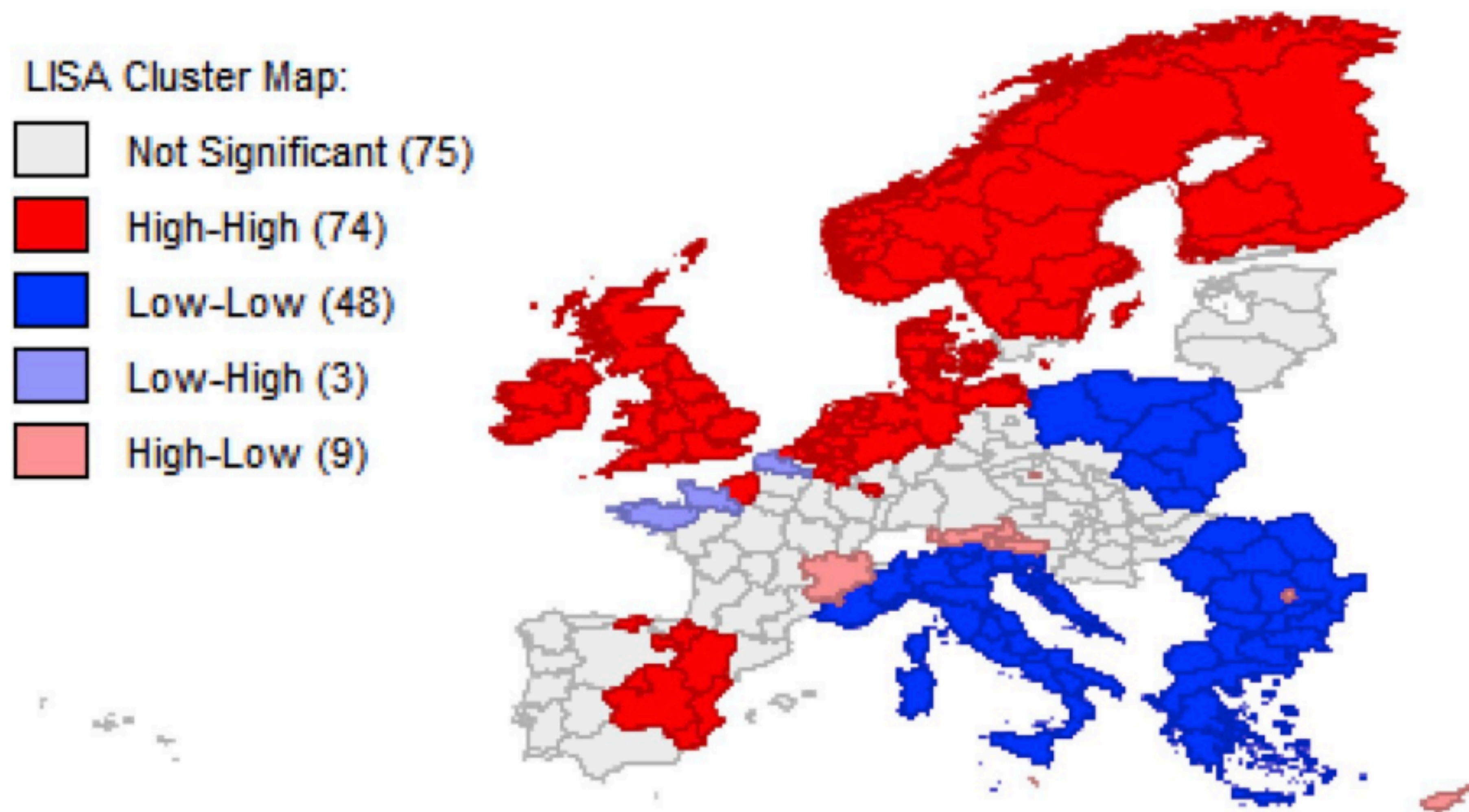


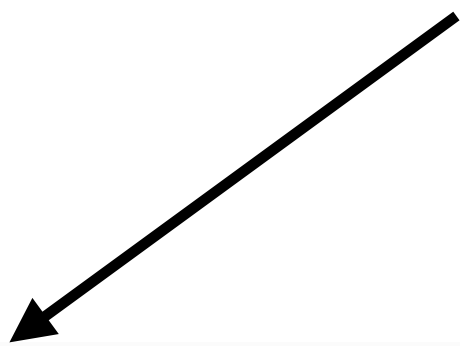
Fig. 12. 2016 proportion of population with on-the-go internet access (n = 209, Moran's I = 0.682).

New library: **esda** (exploratory spatial data analysis)

Part of PySAL family

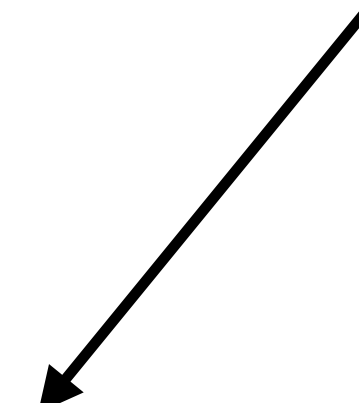
Data to examine for spatial autocorrelation (gdf)

```
mi = esda.Moran(br['Pct_Leave'], w)
```



Spatial weights

```
lisa = esda.Moran_Local(br['Pct_Leave'], w)
```



Jupyter

Project Exam

- Analysis answering a **research question or prototype of digital product**
- Groups (1-3 people) - remember contribution statement
- Hand in report *and* code/product
- LaTeX is recommended (template suggestion on LearnIT)

Detailed instructions on LearnIT

Geospatial Data Science Exam Project Instructions, Spring 2026

The submission is a written project report about the application of geospatial data science either to **answer a research question** or to **create a prototype of a digital product**. It may range from a technical workflow proof of concept to research data exploration. The project should explore or solve a problem with a geospatial dimension and may focus on any aspect of spatial data collection, visualization, analysis, or statistical evaluation. The submission has two parts: (a) commented code deposited on Github (or similar code repository) and (b) the associated report that describes the project and links to the code repository. No use of generative artificial intelligence is allowed at the exam under any circumstances.

<https://learnit.itu.dk/course/section.php?id=168948>

Use the exam resources

Data sets

There are a lot of open and free spatial data sets, so listed below is only a small list of examples. Always remember to be critical of data quality and origin!

[OpenStreetMap](#) (data can be accessed through for example OSMnx or from [Geofabrik](#), [BBike](#) or the Overpass API)

[Opendata.dk](#) (open datasets from Danish municipalities)

[Dataforsyning.dk](#) (official portal for open public spatial data in Denmark. Data are free but you need to create a user to access it)

[Datafordeler.dk](#) (official portal for open public data in Denmark. Data are free but you need to create a user to access it)

<https://geographicdata.science/book/data/README.html>

[Global Human Settlement - GHSL Homepage - European Commission \(europa.eu\)](#)

<https://freegisdata.rtwilson.com/>

<https://geodacenter.github.io/data-and-lab/>

<http://snap.stanford.edu/data/index.html#locnet> (Two data sets of location-based online social networks)

[Åpne data - Oslo Bysyssel](#) (open data on city bikes in Oslo)

Ask questions if you're stuck or in doubt!

<https://learnit.itu.dk/course/section.php?id=168948>

Project Exam

	Group	Members	Contact email	Project name	Project description (what data do you use, what do you do with it)	Approved (by Michael)
Example row	0	Michael Szell Ane Rahbek Vierø	misz@itu.dk	Bicycle crashes London	We will research the spatial properties of bicycle crashes over the year 2021 in London using road crash data from the UK Dept for Transport (from https://data.gov.uk/dataset/cb7ae6f0-4be6-4935-9277-47e5ce24a11f/road-safety-data, last update Nov 25 2021) and street network data from OpenStreetMap. We will study the spatial clustering of crashes, and relate them to variables such as weather, road type, speed limit, pavement, protected bicycle infrastructure. We will explore the spatial relations between these data, to get a better understanding of the factors that could influence or prevent crashes.	YES
Please add your project in the first empty line from top. After you added a new entry, contact Michael at misz@itu.dk for approval						
	Group	Members	Contact email	Project name	Project description (what data do you use, what do you do with it)	Approved (by Michael)
Your rows to fill in:	1					NOT YET
	2					NOT YET
	3					NOT YET
	4					NOT YET
	5					NOT YET
	6					NOT YET

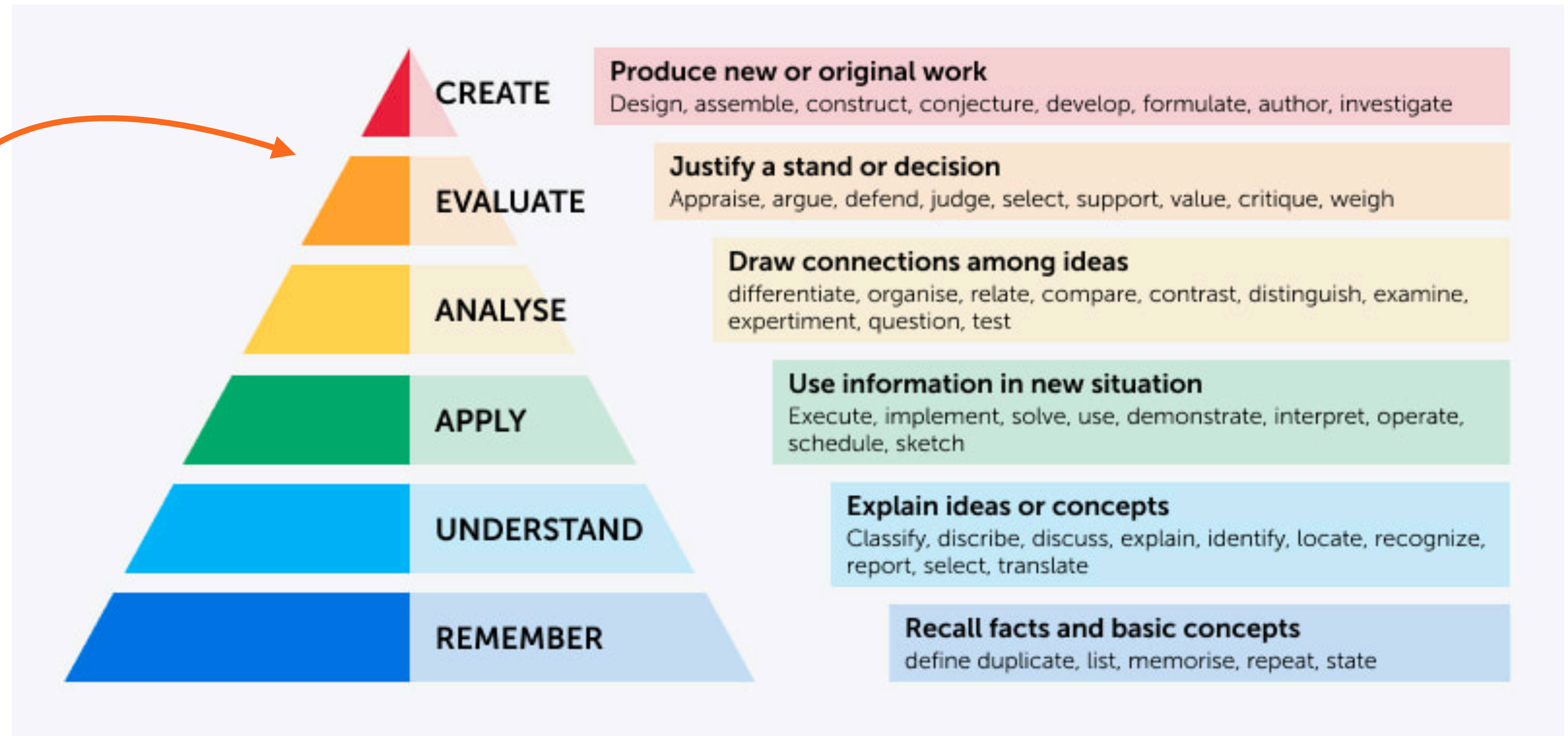
Deadline for project proposals: April 8

Exam hand-in: May 18, 14:00

Intended Learning Outcomes (ILOs)

- **Demonstrate GIS/GDS concepts and be able to use relevant Python libraries** programmatically to import, manipulate and analyze spatial data in different formats. Apply a number of spatial analysis techniques and explain how to interpret the results, in a process of turning data into insights.
- **Reflect on the motivation and inner workings** of the main methodological approaches of GDS, both analytical and visual.
- **Critically evaluate the suitability** of a specific GDS technique, what it can offer and how it can help answer questions of interest.
- **Apply a number of spatial analysis techniques and explain** how to interpret the results, in a process of turning data into insights.
- When faced with a new data-set, work independently using GIS/GDS tools programmatically to **extract valuable insight**.

Bloom's Taxonomy



Better narrow scope and high level, than wide scope and low level!

Sources and further materials for today's class



***Geographic Data Science
with Python***



[https://geographicdata.science/book/notebooks/
06_spatial_autocorrelation.html](https://geographicdata.science/book/notebooks/06_spatial_autocorrelation.html)

[https://geographicdata.science/book/notebooks/
07_local_autocorrelation.html](https://geographicdata.science/book/notebooks/07_local_autocorrelation.html)

https://darribas.org/gds_course/content/bF/concepts_F.html

https://mgimond.github.io/Spatial/chp13_0.html

Next week: Spatial Clustering

